

1 Pseudopotential Kohn–Sham accuracy

We next assess the accuracy of SPARC-atomSFE for pseudopotential Kohn–Sham DFT calculations through comparisons with the literature. In particular, we compare against the spectral scheme based SPARC-atom code, implemented as part of the M-SPARC package, while considering four exchange–correlation functionals: LDA, PBE, rSCAN, and PBE0. We compare the results for seven atoms spanning a range of chemical environments—He, N, O, Fe, Mn, Mo, and Cs. The distribution of the differences in the results is shown as violin plots in Figure 1. We observe that total energies and occupied valence eigenvalues agree to 10^{-6} Ha or better for LDA, PBE, and rSCAN. The exception is PBE0, which exhibits errors approximately an order of magnitude larger; this can be attributed to the fact that the quadrature in the spectral scheme used by SPARC-atom is limited to the degree of the polynomial used, making it difficult to converge calculations involving exact exchange, where higher-order quadrature is required. This limitation can be overcome by adopting a differential approach for the exact exchange operator, as done in the present work. These results demonstrate the accuracy of SPARC-atomSFE for atomic structure calculations based on pseudopotential Kohn–Sham DFT.

As supplementary material to the accuracy discussion above, we tabulate the full numerical data underlying Figure 1 in Tables 1, 2, 3, and 4 (LDA-SVWN, GGA-PBE, rSCAN, and PBE0, respectively).

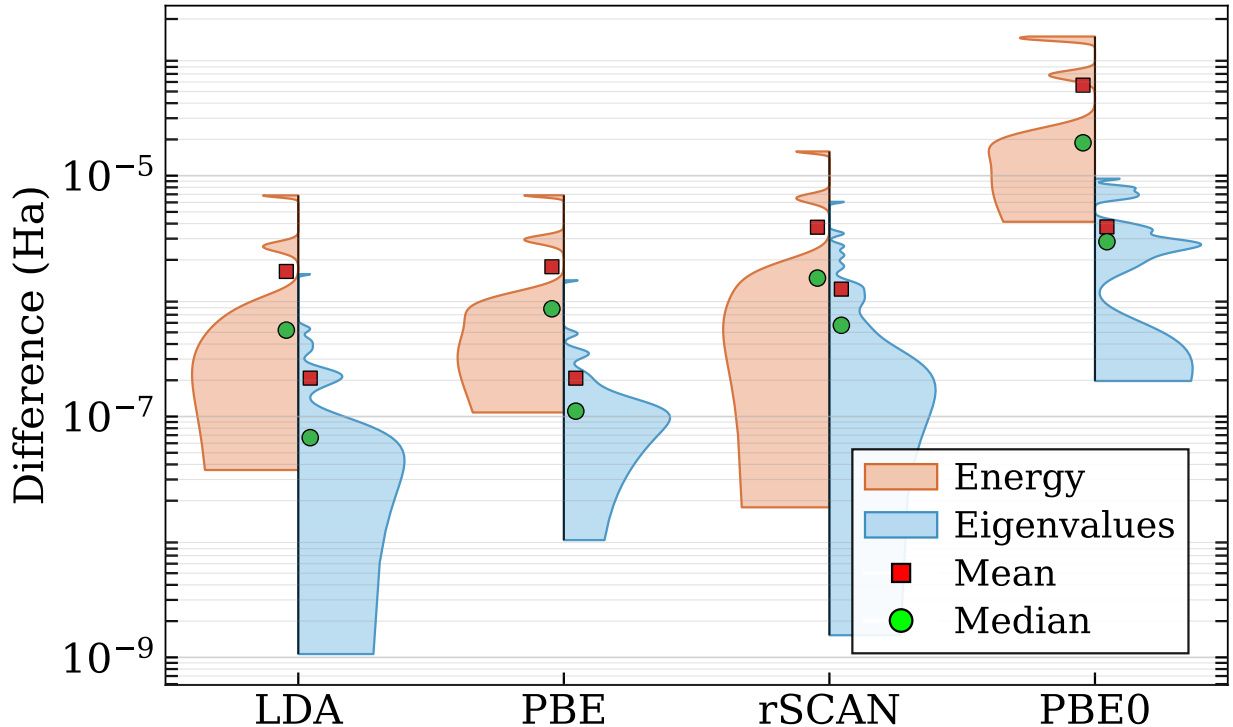


Figure 1: Comparison of SPARC-atomSFE pseudopotential Kohn–Sham DFT results with the M-SPARC SPARC-atom reference for various exchange–correlation functionals. The energy and eigenvalue errors denote the absolute differences in the corresponding quantities.

Table 1: Comparison of the total energies and eigenvalues for the LDA-SVWN exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

	Z	$-E_{\text{tot}}$ (Ha)			$-\varepsilon_{nl}$ (Ha)						
		atomSFE	M-SPARC	Difference	n	ℓ	g_{nl}	atomSFE	M-SPARC	Difference	
He	2	2.800083168	2.800083994	8.26×10^{-7}	1	0	2	0.570459	0.57046	4.23×10^{-7}	
N	7	10.022867763	10.022865183	2.58×10^{-6}	2	0	2	0.677081	0.677081	2.01×10^{-7}	
					2	1	3	0.266124	0.266124	2.31×10^{-7}	
O	8	16.2174808	16.217480836	3.6×10^{-8}	2	0	2	0.873064	0.873065	5.34×10^{-7}	
					2	1	4	0.338133	0.338133	2.05×10^{-7}	
Mn	25	105.23371943	105.233726291	6.86×10^{-6}	3	0	2	3.138754	3.138755	3.56×10^{-7}	
					3	1	6	2.002629	2.002629	2.23×10^{-7}	
					3	2	5	0.25773	0.257732	1.52×10^{-6}	
					4	0	2	0.194097	0.194097	6.97×10^{-8}	
Fe	26	116.74117455	116.741174029	5.22×10^{-7}	3	0	2	3.436694	3.436694	6.19×10^{-8}	
					3	1	6	2.202574	2.202574	5.69×10^{-8}	
					3	2	6	0.285108	0.285108	6.08×10^{-8}	
					4	0	2	0.201428	0.201428	1.31×10^{-8}	
Mo	42	69.690800845	69.690800908	6.32×10^{-8}	4	0	2	2.367637	2.367637	3.26×10^{-8}	
					4	1	6	1.418244	1.418244	7.24×10^{-8}	
					4	2	5	0.144562	0.144562	6.2×10^{-9}	
					5	0	1	0.158758	0.158758	2.1×10^{-9}	
Cs	55	23.310268861	23.310269188	3.27×10^{-7}	5	0	2	0.988689	0.988689	3.19×10^{-8}	
					5	1	6	0.500169	0.500169	6.39×10^{-8}	
					6	0	1	0.081715	0.081715	1.07×10^{-9}	

Table 2: Comparison of the total energies and eigenvalues for the GGA-PBE exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

$-E_{\text{tot}}$ (Ha)					$-\varepsilon_{nl}$ (Ha)					
Z		atomSFE	M-SPARC	Difference	n	ℓ	g_{nl}	atomSFE	M-SPARC	Difference
He	2	2.853330035	2.853330926	8.91×10^{-7}	1	0	2	0.57931	0.579311	3.46×10^{-7}
N	7	10.059303061	10.059300117	2.94×10^{-6}	2	0	2	0.682915	0.682914	1.07×10^{-7}
					2	1	3	0.260551	0.260551	1.12×10^{-7}
O	8	16.273706435	16.273706327	1.08×10^{-7}	2	0	2	0.880576	0.880576	4.88×10^{-7}
					2	1	4	0.331872	0.331872	1.47×10^{-7}
Mn	25	105.366008921	105.366015798	6.88×10^{-6}	3	0	2	3.15601	3.15601	2.33×10^{-7}
					3	1	6	2.006265	2.006265	9.38×10^{-8}
					3	2	5	0.248833	0.248834	1.35×10^{-6}
					4	0	2	0.187687	0.187687	1.98×10^{-7}
Fe	26	116.778840527	116.778840314	2.13×10^{-7}	3	0	2	3.455078	3.455077	1×10^{-7}
					3	1	6	2.206535	2.206535	7.31×10^{-8}
					3	2	6	0.275801	0.275801	5.68×10^{-8}
					4	0	2	0.194482	0.194482	1.64×10^{-7}
Mo	42	69.679429583	69.679429125	4.58×10^{-7}	4	0	2	2.364713	2.364713	4.52×10^{-8}
					4	1	6	1.414321	1.414321	6.31×10^{-8}
					4	2	5	0.137922	0.137922	1.1×10^{-7}
					5	0	1	0.150181	0.150181	1.19×10^{-7}
Cs	55	23.249841526	23.24984231	7.84×10^{-7}	5	0	2	0.982389	0.982389	2.49×10^{-8}
					5	1	6	0.496788	0.496788	9.42×10^{-9}
					6	0	1	0.07667	0.076669	3.2×10^{-7}

Table 3: Comparison of the total energies and eigenvalues for the rSCAN exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

$-E_{\text{tot}}$ (Ha)					$-\varepsilon_{nl}$ (Ha)					
	Z	atomSFE	M-SPARC	Difference	n	ℓ	g_{nl}	atomSFE	M-SPARC	Difference
He	2	2.803317372	2.803317354	1.76×10^{-8}	1	0	2	0.605361	0.605361	7.91×10^{-8}
N	7	9.620390196	9.620390518	3.22×10^{-7}	2	0	2	0.729363	0.729363	1.53×10^{-7}
					2	1	3	0.265639	0.265639	9.57×10^{-8}
O	8	15.701679072	15.701680484	1.41×10^{-6}	2	0	2	0.940105	0.940105	9.16×10^{-7}
					2	1	4	0.343354	0.343355	1.19×10^{-6}
Mn	25	99.176114883	99.176113426	1.46×10^{-6}	3	0	2	3.279161	3.279161	2.91×10^{-7}
					3	1	6	2.085219	2.085219	2.29×10^{-7}
					3	2	5	0.251825	0.251825	3.51×10^{-7}
					4	0	2	0.195917	0.195917	1.53×10^{-9}
Fe	26	118.003919473	118.003912979	6.49×10^{-6}	3	0	2	3.585989	3.585987	2.6×10^{-6}
					3	1	6	2.293534	2.293532	2.19×10^{-6}
					3	2	6	0.281913	0.281911	1.76×10^{-6}
					4	0	2	0.203008	0.203008	5.85×10^{-7}
Mo	42	68.167004207	68.16700474	5.33×10^{-7}	4	0	2	2.45666	2.456659	9.19×10^{-7}
					4	1	6	1.469806	1.469806	5.61×10^{-7}
					4	2	5	0.13904	0.13904	1.74×10^{-7}
					5	0	1	0.154164	0.154164	1.45×10^{-7}
Cs	55	20.179672075	20.179656185	1.59×10^{-5}	5	0	2	1.023123	1.023124	1.23×10^{-6}
					5	1	6	0.518867	0.51887	3.31×10^{-6}
					6	0	1	0.076443	0.076437	6.07×10^{-6}

Table 4: Comparison of the total energies and eigenvalues for the PBE0 exchange-correlation functional based on norm-conserving pseudopotential atomic calculations.

$-E_{\text{tot}}$ (Ha)					$-\varepsilon_{nl}$ (Ha)					
Z		atomSFE	M-SPARC	Difference	n	ℓ	g_{nl}	atomSFE	M-SPARC	Difference
He	2	2.858442869	2.858447011	4.14×10^{-6}	1	0	2	0.668802	0.668805	2.7×10^{-6}
N	7	9.840929774	9.840936922	7.15×10^{-6}	2	0	2	0.781192	0.781194	2.04×10^{-6}
					2	1	3	0.268917	0.268919	1.68×10^{-6}
O	8	16.010353526	16.010372278	1.88×10^{-5}	2	0	2	0.999295	0.999299	3.58×10^{-6}
					2	1	4	0.36414	0.364142	2.64×10^{-6}
Mn	25	103.753694182	103.753829602	1.35×10^{-4}	3	0	2	3.363098	3.363104	6.46×10^{-6}
					3	1	6	2.162649	2.162657	8.15×10^{-6}
					3	2	5	0.258814	0.258822	7.74×10^{-6}
					4	0	2	0.214787	0.214786	3.11×10^{-7}
Fe	26	115.71864899	115.718792042	1.43×10^{-4}	3	0	2	3.665564	3.665571	6.87×10^{-6}
					3	1	6	2.363997	2.364006	9.45×10^{-6}
					3	2	6	0.298827	0.298834	7.26×10^{-6}
					4	0	2	0.222093	0.222093	2.59×10^{-7}
Mo	42	68.967998129	68.968066473	6.83×10^{-5}	4	0	2	2.526861	2.526864	3.48×10^{-6}
					4	1	6	1.53111	1.531114	4.05×10^{-6}
					4	2	5	0.144336	0.144338	2.35×10^{-6}
					5	0	1	0.147808	0.147807	1.97×10^{-7}
Cs	55	22.420055553	22.420073893	1.83×10^{-5}	5	0	2	1.066609	1.066612	2.94×10^{-6}
					5	1	6	0.551291	0.551294	2.72×10^{-6}
					6	0	1	0.076059	0.076059	2.34×10^{-7}